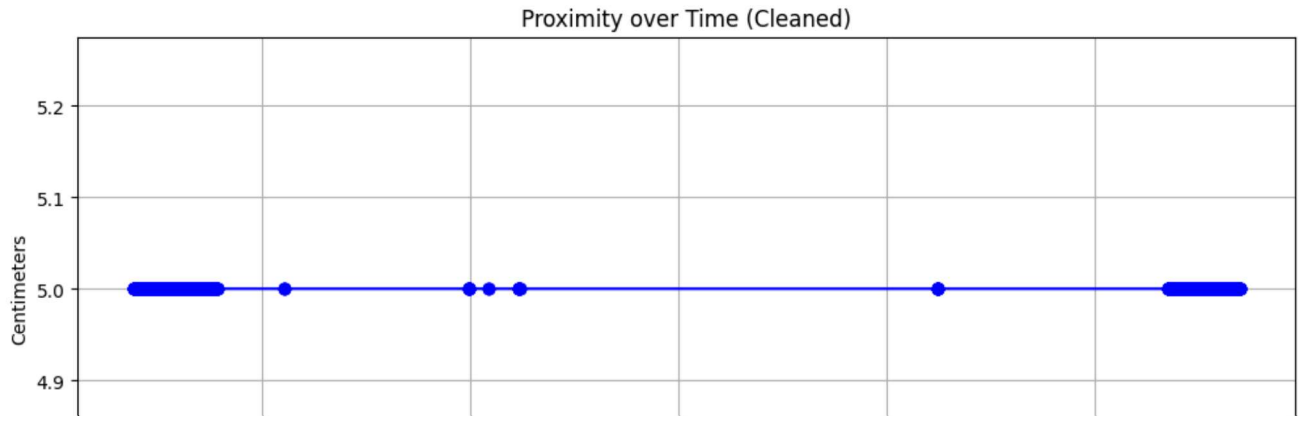
```
import pandas as pd
import io
import matplotlib.pyplot as plt


df = pd.read_csv("/content/Proximity.csv")

print("--- Initial Info ---")
print(df.info())

# 2. Clean Duplicates

df = df.drop_duplicates().reset_index(drop=True)


df['Datetime'] = pd.to_datetime(df['Timestamp']) + pd.to_timedelta(df['Milliseconds'], unit='ms')


df = df.sort_values(by='Datetime')
df.set_index('Datetime', inplace=True)



# Most of your centimeter data is exactly 5.000305.
# We'll calculate the 'change' to make it easier to see when things actually move.
df['delta_cm'] = df['centimeters'].diff()

# Drop columns that are now redundant
df.drop(columns=['Timestamp', 'Milliseconds'], inplace=True)

print("\n--- Preprocessed Data (Head) ---")
print(df.head())

# 8. Quick Visualization
plt.figure(figsize=(12, 5))
plt.plot(df.index, df['centimeters'], marker='o', linestyle='-', color='b')
plt.title('Proximity over Time (Cleaned)')
plt.xlabel('Time')
plt.ylabel('Centimeters')
plt.grid(True)
plt.show()
```

```
--- Initial Info ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7970 entries, 0 to 7969
Data columns (total 3 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Timestamp     7970 non-null   object
 1   Milliseconds  7970 non-null   int64
 2   centimeters   7970 non-null   float64
dtypes: float64(1), int64(1), object(1)
memory usage: 186.9+ KB
None

--- Preprocessed Data (Head) ---
                        centimeters  delta_cm
Datetime
2020-05-25 21:26:55.002     5.000305       NaN
2020-05-25 21:26:55.101     5.000305       0.0
2020-05-25 21:26:55.201     5.000305       0.0
2020-05-25 21:26:56.301     5.000305       0.0
2020-05-25 21:26:56.402     5.000305       0.0
```